

JOONWON LEE

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SUMMARY

Statistics Ph.D. candidate and quantitative researcher specializing in probabilistic modeling, scalable likelihood-based inference, hypothesis testing, simulation, and signal-versus-noise diagnostics. Builds reproducible Python/PyTorch research frameworks for large, complex datasets, rapidly tests empirical ideas, and adapts models when evidence reveals noise, misspecification, or changing dependence. Brings hands-on market-making strategy experience and a strong interest in research-driven trading.

EDUCATION

Ph.D. in Statistics, Rutgers University, Piscataway, NJ

Expected May 2027 | GPA: 3.8/4.0

M.S. in Statistics, University of Minnesota, Minneapolis, MN

Sep 2021 | GPA: 3.7/4.0

TECHNICAL SKILLS

Programming and Research Systems: Python (NumPy, Pandas, SciPy, PyTorch, LightGBM, scikit-learn), C++/pybind11 integration, SQL, Linux, Git, AWS EC2, HPC/SLURM; CPU/GPU computing, cached artifacts, and restartable analytical pipelines

Quantitative Research: Probability and stochastic processes, hypothesis testing, MLE/GLS, Monte Carlo simulation, numerical optimization, Gaussian processes, time-series and spatiotemporal modeling, regression and machine learning, spectral analysis, model selection, validation, and sensitivity analysis

EXPERIENCE

JPMorgan Chase - Summer Quantitative Analytics Associate, Model Risk Governance and Review

Summer 2026

- Developed and validated a Sequential Probability Ratio Test for binary risk indicators, converting a monitoring problem into explicit hypotheses, likelihood-ratio boundaries, and controlled sequential decisions.
- Combined Brownian-motion approximation, dynamic-programming probability propagation, and simulation to evaluate boundary-crossing probabilities, stopping behavior, Type I/II error, and early-decision tradeoffs.

Travelers - Data Science Leadership Program, Business Insurance Pricing Team

Summer 2025

- Developed an end-to-end LightGBM pipeline across 2.46M+ records, including data preparation, feature processing, GLM benchmarking, model selection, validation, and performance diagnostics on AWS EC2.
- Evaluated model performance and feature-attribution patterns and translated findings into pricing and risk-segmentation recommendations for business stakeholders.

QUANTITATIVE RESEARCH AND TRADING

IMC Prosperity Algorithmic Trading Competition - Top 2.39% of Participants

Apr 2025

- Built and evaluated a market-making strategy using market-based fair-value estimation and inventory-aware quoting.
- Implemented rolling-window exposure controls that penalized persistent inventory and triggered soft/hard liquidation to limit position risk.

Ph.D. Dissertation Research - Scalable Probabilistic Modeling and Diagnostics

Sep 2024 - Present

- Probabilistic Modeling:** Developed an advection-aware Vecchia likelihood approximation for nonseparable Gaussian processes, using ordered conditional models to fit up to 145,008 observations per day.
- Signal-versus-Noise Diagnostics:** Designed simulation studies and scale- and frequency-resolved tools to determine whether apparent structure reflects persistent dependence, noise, missingness, or model misspecification.
- Research Framework:** Built reusable Python/PyTorch workflows for data-quality filtering, missingness tracking, time-dependent alignment, estimation, simulation, cached outputs, and restartable CPU/GPU and HPC execution.
- Performance Engineering:** Integrated compiled C++ max-min ordering routines into the Python research pipeline through pybind11 for performance-critical dependency ordering.

ADDITIONAL

Coursework: Probability Theory, Stochastic Processes, Machine Learning, Linear Algebra, Statistical Computing, Advanced Theory of Statistics I-II, Data Structures and Algorithms, Econometrics

Languages: Korean (Native); English (Fluent); Chinese (Basic)